# How to not write parsers

Wow, that’s a lot of grammars, too bad I’m not reading them

Curtis Chin Jen Sem

Scan for these slides

# It started with wanting `.cabal` files to look nice

- syntax-highlighting is nice.
- started with wanting it for `.cabal` files in my editor helix.
  - pre-existing `.cabal` tree-sitter grammar by Magnus Therning

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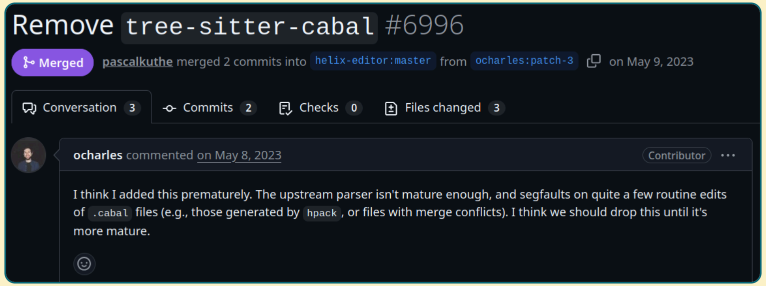

# It started with wanting `.cabal` files to look nice

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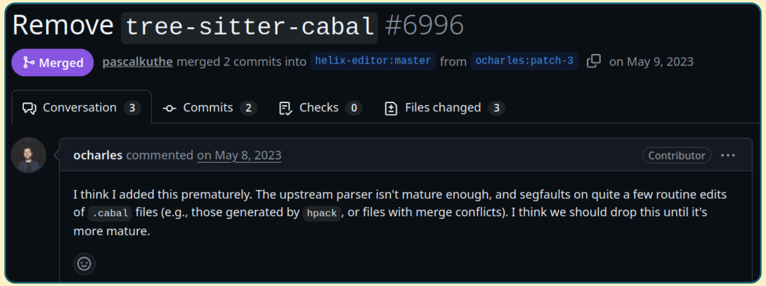


- a PR by Ananda Umamil fixed the segfaults
- but then I wanted `cabal.project`, which look like `.cabal` files
- Grammar?

# Is this a valid `.cabal` file?

- a grammar is just the set of files that count. so: is this one in it?

```
Library {
    build-depends: base ≥ 4.0 && < 4.2,
                   syb  ≥ 0.1 && < 0.2

    if impl(ghc < 6.9) {
        buildable: False
    }
}
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cabal says **valid, no warnings**

- lots of legacy syntax
- there is a whole brace-delimited syntax

# How about this one?

```
library
    build-depends: base

-- Duplicating inclusion plugin conditions
    if flag(eval)
        cpp-options: -Dhls_eval
```

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- a comment at column zero does not end the stanza
- the `if` still belongs to the `library` above it
- in `haskell-language-server.cabal`

# And these? Which of these is right?

```
library
  if os(windows)
    build-depends: Win32
  elif os(linux)
    build-depends: unix
```

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  if os(windows)
    build-depends: Win32
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# And these? Which of these is right?

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library
  if os(windows)
    build-depends: Win32
  elif os(linux)
    build-depends: unix
```

`cabal` **valid, since `cabal-version: 2.2`**

```
library
  if os(windows)
    build-depends: Win32
  elseif os(linux)
    build-depends: unix
```

`cabal` ***invalid subsection*, ignored**

- `elif` is the real one.
- `elseif` still parses, warns, and drops the branch
- both successfully `cabal build`!

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- `.cabal` is twenty years of accumulated legacy
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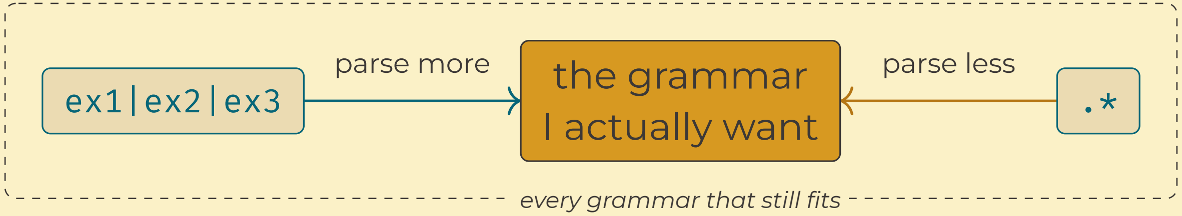
- `.cabal` is twenty years of accumulated legacy
- writing parsers are already hard
- coverage is a problem
- so what if I don’t?
- instead of a parsing problem,
- … how about tackling it like a constraint solving problem?



# Tree-sitter

- a parser generator DSL for incremental parsers
- grammar written in javascript, generates parsers in C
- out comes a concrete syntax tree, covering every byte of the file

```
library
    build-depends: base ⩾ 4.9
```

```
(library
  type: (section_type)
  properties: (property_or_conditional_block
    (field
      name: (field_name)
      value: (field_value
        (identifier)
        (constraint_op)
        (version)))))
```

# Tree-sitter

- always returns a tree even if syntax is invalid

```
library
    build-depends base ⩾ 4.9
                  ↑ no colon
```

```
(library
  type: (section_type))
(ERROR
  (field_name)
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# Tree-sitter

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```

- most common usecase is syntax highlighting
- editors have to highlight files that are actively edited
- also used for tags, tree-based navigation, code-folding

# Pattern matching parse trees

```
library core
    build-depends: base
```

its parse tree

```
(cabal
  (sections
    (library
      type: (section_type)
      name: (section_name)
      properties: (property_or_conditional_block
        (field
          name: (field_name)
          value: (field_value
            (identifier)))))))
```

query

```
(library

  name: (section_name) @name)
```

# Pattern matching parse trees

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library core
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            (identifier)))))))
```

query

```
(library

  name: (section_name) @name)
```

@name captures **`core`**

- change the capture to `type: (section_type)` and you get `library`

# Using captures

- going from captures to behavior

pkg.cabal, three stanzas:

```
library core
    build-depends: base
executable mypkg
    ghc-options: -O2
flag fast
    default: False
```

tags.scm, one pattern per symbol kind

```
(library     ...  @definition.module
(executable  ...  @definition.function
(flag        ...  @definition.constant
 ... is  name: (section_name) @name)
```

what a tags reader gets

```
→ core  module
→ mypkg  function
→ fast  constant
```

- the outer node decides which stanza matches
- `@name` gives the symbol's text, `definition.<kind>` gives its kind
- so `tags.scm` turns node names into definition variants

# Features as queries

- pull nodes out of the tree, tag each one with a capture

| query file | pulls out of the tree | which Helix exposes as | lines |
|---|---|---|---|
| `highlights.scm` | every literal, name, operator | colour | 101 |
| `textobjects.scm` | sections and their bodies | select or jump a stanza | 45 |
| `indents.scm` | section and `if` bodies | where the cursor goes on enter | 32 |
| `tags.scm` | section names | the symbol picker | 15 |
| `locals.scm` | `flag` defs and their uses | go-to-definition on a flag | 11 |
| `rainbows.scm` | parens in conditions | bracket pair colours | 7 |

- query one single tree six different ways easily
- only have to give named nodes

# Can the queries even compile?

- every pattern in every query file
- resolved against the generated parser

| the pattern | what `check-queries` says |
| --- | --- |
| `(library name: (section_name) @name)` | captures `core` |
| `(librari name: (section_name) @name)` | `Invalid node type "librari"` |
| `(library nmae: (section_name) @name)` | `Invalid field name "nmae"` |

- node types and field names are checked
- the grammar is now forced to support every query

# Harvesting corpora

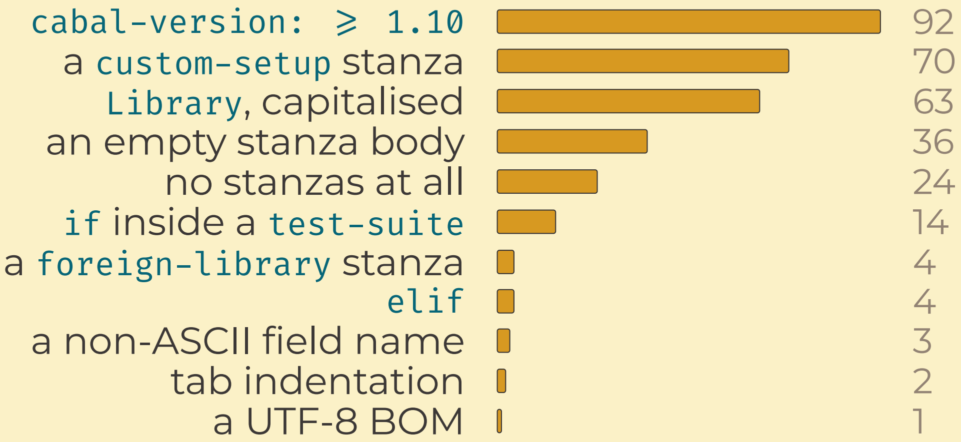
- nobody has written cabal's grammar down
- …but, there's an elaborate testsuite
- `find` over two testsuites, cabal and HLS

| harvested from | files |
|---|---|
| `cabal-testsuite/PackageTests` | 760 |
| `Cabal-tests/…/ParserTests` | 84 |
| `haskell-language-server/test/testdata` | 64 |
| `cabal-install/tests` | 35 |
| the two projects' `.cabal` files | 24 |

- 943/967 test-cases, median 14 lines

# Does every real file parse?

- `parse-corpus` reads all 967 harvested files
- asserts zero `ERROR`, zero `MISSING`
- `tree-sitter-cabal` failed 237



# What did I capture?

- `extract-golden` records capture, span and text over fixtures
- and diffs that against a committed set

```
library core
```

| | | | |
|---|---|---|---|
| the capture I asked for | `name` | `3,8-3,12` | `core` |
| one node to the left | `name` | `3,0-3,7` | `library` |

- the capture slid to the wrong node
- span and text both wrong

# What actually reaches the screen?

- `highlight-golden` records the winning capture
- one row per token, from rendered output

the same `.`, in two fields

```
hs-source-dirs: .    ln 26  string.special.path
 description: … .    ln  8  string
```

# What actually reaches the screen?

- `highlight-golden` records the winning capture
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the same ., in two fields

```
hs-source-dirs: .   ln 26  string.special.path
 description: … .   ln  8  string
```

- one node type interpreted two different ways
- swap those patterns in the file
- and the result changes

# `parse` never fails

```
readFields :: ByteString → Either ParseError [Field]    -- Cabal
parse      :: Text       → Tree                         -- tree-sitter
```

```
library
    build-depends base ⩾ 4.9
                  ↑ no colon
```

*parses to →*

```
(library
  type: (section_type))
(ERROR
  (field_name)
  (identifier)
  (constraint_op)
  (version))
```

did it parse? always yes

`parse-corpus` constrains

# `match` never chooses

```
match :: Query → Tree → [Match]
```

```
                        →   string
hs-source-dirs: .
                        →   string.special.path
```

`match` returns everything

`extract-golden` and `highlight-golden` constrain

# What’s my variable?

each constraint is a fold

constraints $C_1, ..., C_n$, each form a fold from `Tree` into some $M_i$, with expected values $e_i \in M_i$ over fixtures $X$

$$C_i(\text{parse}_{\boldsymbol{g}}\ x) = e_i(x), \quad \forall x \in X$$

solve for $\boldsymbol{g}$, the grammar and the queries

- the constraints are the spec
- every entry is a sample
- the grammar is whatever fits

# Tree constraints can be phrased as `foldMap`

```
foldMap :: Monoid m ⇒ (a → m) → [a] → m
```

*constrain queries by grammar*

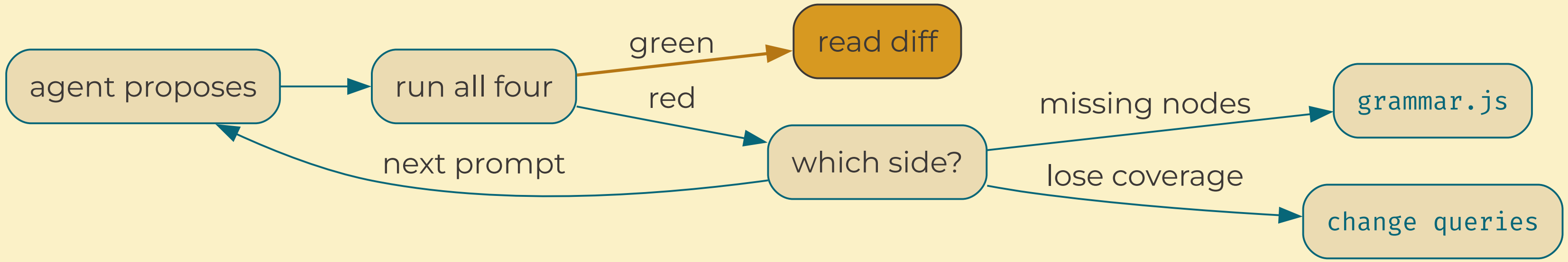
| | | | |
|---|---|---|---|
| $C_1$ | `check-queries` | `queries` ⊆ `grammar` | assertion |

*constrain $g$ over files*

| | | | | | | |
|---|---|---|---|---|---|---|
| $C_2$ | `parse-corpus` | `foldMap all . parse` | assertion | | | |
| $C_3$ | `extract-golden` | `foldMap id . match . parse` | `value.path` | `25,18-25,19` | `'.'` | per capture |
| $C_4$ | `highlight-golden` | `foldMap last . match . parse` | `26` | `string.special.path` | `'.'` | per token |

turn these `monoid` values into tests by rendering

# Both paths are the same job for the agent



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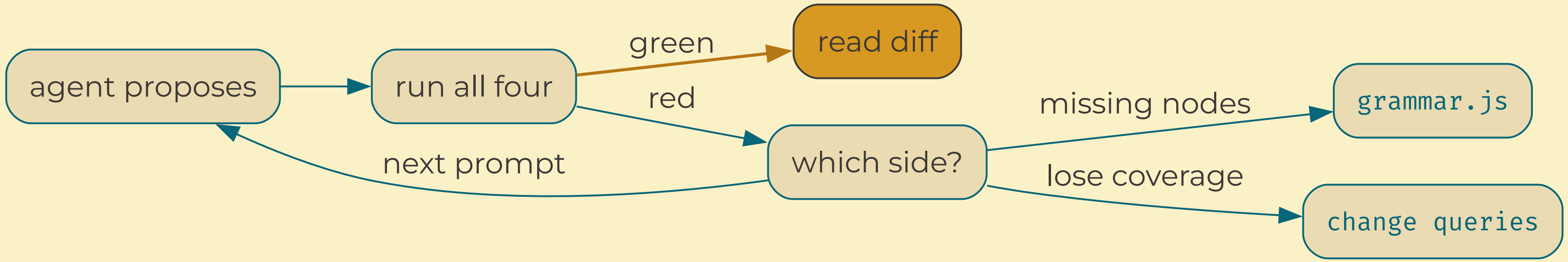
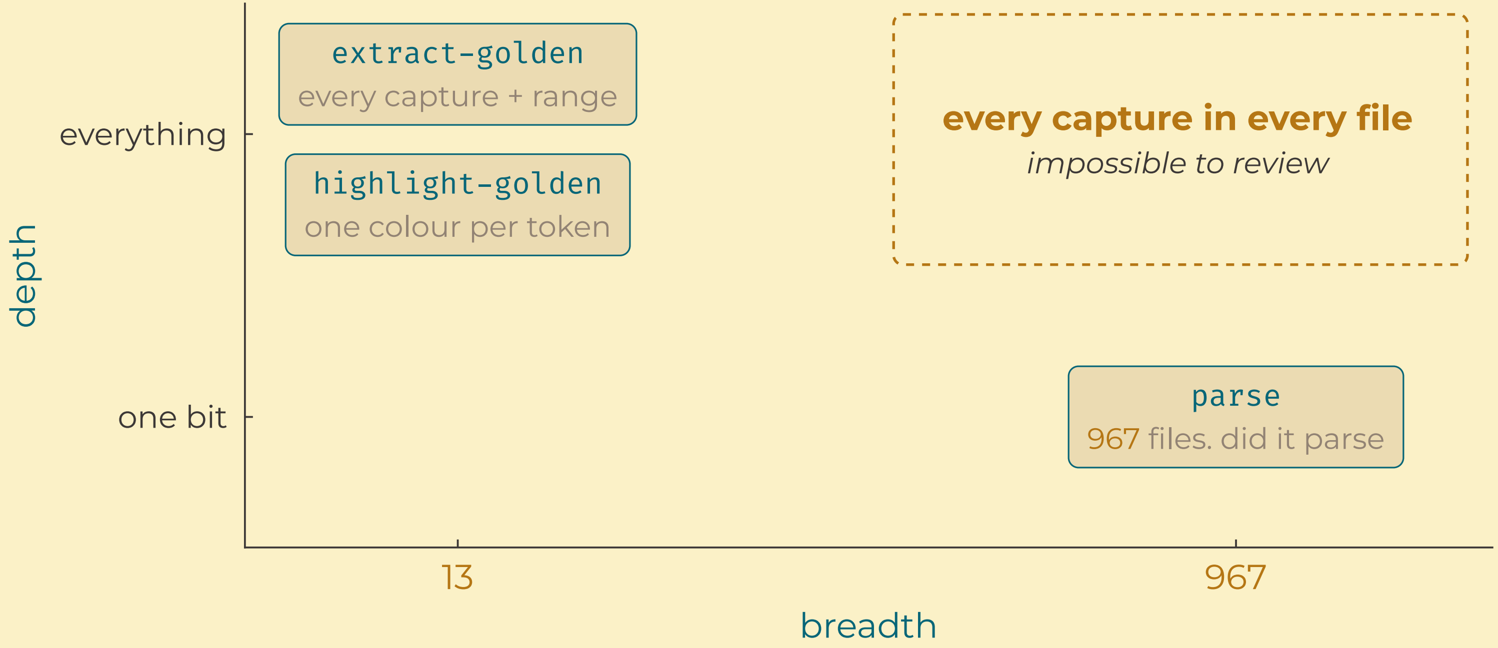



measure, edit, re-run

- only the grammar and queries change
- manual step when everything is correct
- check is this the change I wanted?

# Constraints require different sets




| | |
|---|---|
| breadth | how many files a gate ever reads |
| depth | how much of one file it checks |

stay with what's reviewable

# I didn't write the parser
# **just read everything else**

---

**the agent wrote**
7 grammars, 4 C scanners

**from**
4 constraints, and 2 corpora

*967 cabal files, zero `ERROR` nodes*

*same four gates built GHC's `Core`. nothing to fork, nothing to harvest*

# Questions?

github.com/crtschin/tree-sitter-haskell-contrib

grammars for `.cabal`, `cabal.project`, and GHC's Core / STG / Cmm dumps.

# Appendix

- a format with no grammar and no corpus
- captures vs the screen
- collapsing the two gates into one

# A Core dump is a point in a space I can walk

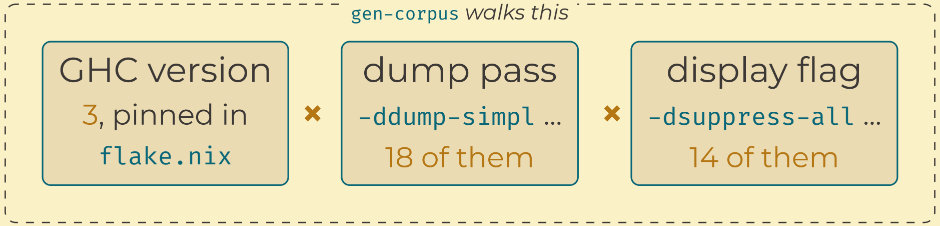
- cabal had an existing grammar to fork
- …and a testsuite I could harvest

a format with no grammar and no corpus

# A Core dump is a point in a space I can walk

- cabal had an existing grammar to fork
- ...and a testsuite I could harvest

- Core has neither, ...so create one
- `gen-corpus` walks GHC's dump parameter space
- same four constraints as cabal




‹ appendix

# It worked here too, and the gates are why

- seven grammars, four C scanners, ~1,300 lines of C
- zero `ERROR` untested nodes
- CI re-parses on every push
- covers multiple GHC versions
- automatically updated with pins

`9.14` moved the pretty-printer
version bumps rot grammars

**this one tripped a test instead**

# one tree, two records

- `extract-golden` records one row per capture
- `highlight-golden` records one row per token

two colour bugs, seen by each record

| | wrong colour | no colour | |
|---|---|---|---|
| per capture | both captures | no capture | blind to colour |
| per token | the colour flips | a grey token | sees the colour |

a colour lives on a token, so the record needs one row per token collapsing the two gates into one

# `extract-golden` could cover `highlight-golden`

```
   extract-golden   foldMap id   . match . parse
 highlight-golden   foldMap last . match . parse
```

| | | |
|---|---|---|
| `extract-golden` | `foldMap id . match . parse` | keep the list |
| `highlight-golden` | `foldMap` **`last`** `. match . parse` | keep the winner |

that whole difference, as one equation over the target:

```
(c on r)  <>  (c' on r)   =   (c' on r)
```

two captures on one range collapse to the later one

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(c on r)  <>  (c' on r)  =  (c' on r)
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two captures on one range collapse to the later one

- highlight is computable from extraction plus a resolver
  - the free monoid is universal, so every reader factors through it
  - I could have run one gate, not two
- that would defeat the point
  - you want different query files to cover different constraints